

Dado :

```
const int buttonPin = 2;
```

```
const int leds[] = {6, 7, 8, 9, 10, 11, 12};
```

```
void setup() {
```

```
    randomSeed(analogRead(0));
```

```
    for (int i = 0; i < 7; i++) {
```

```
        pinMode(leds[i], OUTPUT);
```

```
    }
```

```
    pinMode(buttonPin, INPUT);
```

```
}
```

```
void loop() {
```

```
    if (digitalRead(buttonPin) == HIGH) {
```

```
        sorteia();
```

```
        delay(3000);
```

```
        limpaLeds();
```

```
    }
```

```
}
```

```
void sorteia() {
```

```
    limpaLeds();
```

```
    int valor = random(1, 7);
```

```
    if (valor >= 1) {
```

```

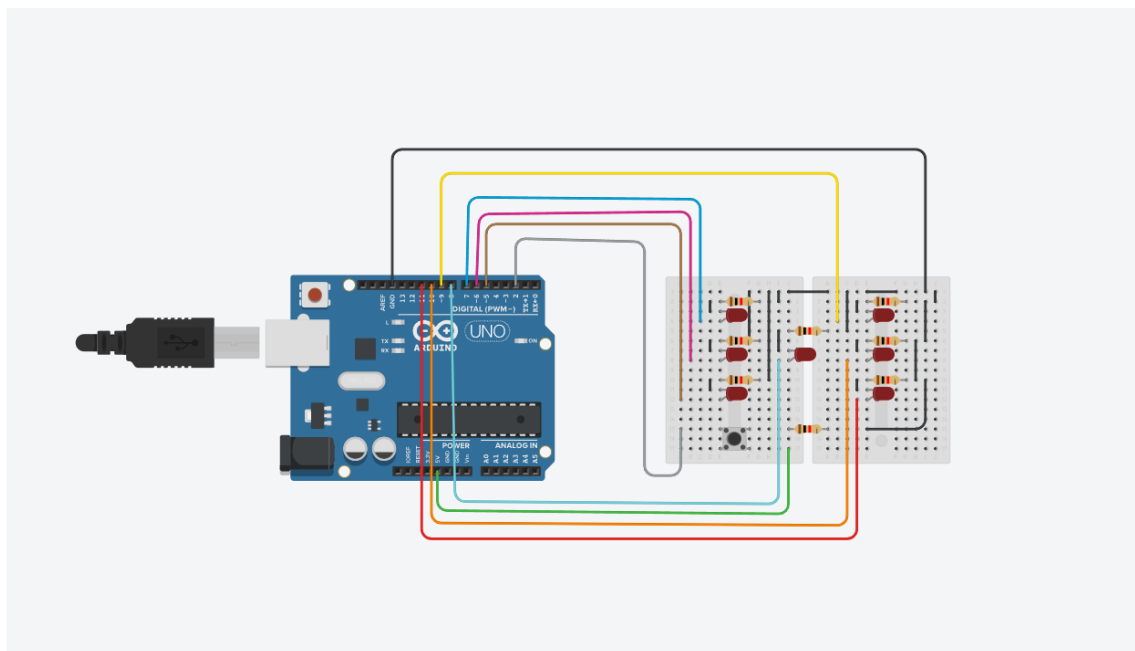
for (int i = 0; i < valor; i++) {
    digitalWrite(leds[i], HIGH);
}
}
}

```

```

void limpaLeds() {
    for (int i = 0; i < 7; i++) {
        digitalWrite(leds[i], LOW);
    }
}

```



Semáforo:

```
int v1_vermelho = 13;
```

```
int v1_amarelo = 12;
```

```
int v1_verde = 11;
```

```
int v2_vermelho = 7;
```

```
int v2_amarelo = 6;
```

```
int v2_verde = 5;
```

```
int ped_vermelho = 4;
```

```
int ped_verde = 3;
```

```
int botao = 2;
```

```
void setup() {
```

```
    pinMode(v1_vermelho, OUTPUT);
```

```
    pinMode(v1_amarelo, OUTPUT);
```

```
    pinMode(v1_verde, OUTPUT);
```

```
    pinMode(v2_vermelho, OUTPUT);
```

```
    pinMode(v2_amarelo, OUTPUT);
```

```
    pinMode(v2_verde, OUTPUT);
```

```
    pinMode(ped_vermelho, OUTPUT);
```

```
    pinMode(ped_verde, OUTPUT);
```

```
    pinMode(botao, INPUT);
```

```
}
```

```
void loop() {

    digitalWrite(v1_verde, HIGH);
    digitalWrite(v1_amarelo, LOW);
    digitalWrite(v1_vermelho, LOW);

    digitalWrite(v2_verde, LOW);
    digitalWrite(v2_amarelo, LOW);
    digitalWrite(v2_vermelho, HIGH);

    digitalWrite(ped_verde, LOW);
    digitalWrite(ped_vermelho, HIGH);

    delay(5000);

    digitalWrite(v1_verde, LOW);
    digitalWrite(v1_amarelo, HIGH);
    delay(2000);

    digitalWrite(v1_amarelo, LOW);
    digitalWrite(v1_vermelho, HIGH);

    if (digitalRead(botao) == HIGH) {
        digitalWrite(ped_vermelho, LOW);
        digitalWrite(ped_verde, HIGH);
        delay(4000);

        digitalWrite(ped_verde, LOW);
```

```
digitalWrite(ped_vermelho, HIGH); delay(500);  
digitalWrite(ped_vermelho, LOW); delay(500);  
digitalWrite(ped_vermelho, HIGH); delay(500);  
digitalWrite(ped_vermelho, LOW); delay(500);  
  
digitalWrite(ped_vermelho, HIGH);  
}
```

```
digitalWrite(v1_verde, LOW);  
digitalWrite(v1_amarelo, LOW);  
digitalWrite(v1_vermelho, HIGH);
```

```
digitalWrite(v2_verde, HIGH);  
digitalWrite(v2_amarelo, LOW);  
digitalWrite(v2_vermelho, LOW);
```

```
delay(5000);  
digitalWrite(v2_verde, LOW);  
digitalWrite(v2_amarelo, HIGH);  
delay(2000);
```

```
nda chance do botão) ---  
digitalWrite(v2_amarelo, LOW);  
digitalWrite(v2_vermelho, HIGH);
```

```
if (digitalRead(botao) == HIGH) {  
    digitalWrite(ped_vermelho, LOW);  
    digitalWrite(ped_verde, HIGH);  
    delay(4000);
```

```
digitalWrite(ped_verde, LOW);
```

```
digitalWrite(ped_vermelho, HIGH); delay(500);  
digitalWrite(ped_vermelho, LOW); delay(500);  
digitalWrite(ped_vermelho, HIGH); delay(500);  
digitalWrite(ped_vermelho, LOW); delay(500);
```

```
digitalWrite(ped_vermelho, HIGH);  
}  
}
```

